Selective Test-Time Adaptation and Fail-Closed Safeguards for Digital Pathology: Multicenter Classification and Dense Detection Under Domain Shift

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***Abstract*—Deep learning models trained on source pathology centers frequently degrade when deployed across unseen hospitals and optical scanners. Test-time adaptation (TTA) aims to mitigate distribution shift using unlabeled target data, but unconstrained entropy minimization is unstable: under a fair BatchNorm-affine protocol on Camelyon17-WILDS patch metastasis classification, Tent collapses on 2 of 5 random seeds ($F_1 \approx 0.05$). We investigate selective TTA and fail-closed safeguards across two distinct histopathology tasks: binary metastasis classification and dense mitotic figure detection. For patch classification on Camelyon17-WILDS (ResNet-18, target Hospital 2, $N = 85{,}054$ patches across 10 whole-slide images), a hard uncertainty gate admits target patches to adaptation only when predictive uncertainty falls below a threshold τ fixed on a held-out center. Across five random seeds, selective gating reduces the harmful adaptation rate from 0.187 ± 0.179 (Tent) to 0.007 ± 0.003 ($> 25\times$ reduction) and improves F_1 score on every seed ($+0.023$ vs. frozen source, 95% CI $[0.010, 0.036]$). However, Monte Carlo Dropout variance (UTTA-MED) achieves similar observed performance to softmax confidence gating ($p = 0.869$), and AUROC gains are not statistically significant ($+0.006$, 95% CI $[-0.012, +0.024]$). To resolve single-class attractor collapse on stain outliers, we formulate UTTA-MED coupling Entropic Optimal Transport prototype alignment (OT-TTA) with fail-closed runtime safeguards (SafeAdapt). On Camelyon17 Hospital 2, OT-TTA mathematically precludes majority-class attractor collapse, reversing negative transfer on Slide 29 ($0.8592 \rightarrow 0.9324 \pm 0.0078$ mean AUROC; 0.9360 on seed 42) and elevating cohort performance to 0.9495 ± 0.0134 macro-AUROC across five random seeds ($p = 0.0312$), cutting Expected Calibration Error to 0.0745, and slashing harmful adaptation to 0.0018 ± 0.0003. For dense mitotic figure detection on held-out MIDOG 2021 Scanner 3 (Faster R-CNN, 50 cases, 1,100 spatially stratified patches, 3 seeds), we introduce the Cellularity Mitosis Ratio (CMR), which conditions silence monitoring on hematoxylin optical density, eliminating all 14 false silence alarms on benign acellular stroma and enabling continuous active adaptation ($F_1 = 0.5160 \pm 0.0050$, $\Delta F_1 = +0.0575$, $p < 0.001$). Finally, we prove a PAC L_∞ drift bound guaranteeing that parameter projection ($\|\theta_t - \theta_0\|_\infty \leq 0.05$) bounds maximum output probability displacement to ≤ 0.0460 and enforce episodic whole-slide deterministic resets ensuring patient diagnostic isolation.**

***Index Terms*—Test-time adaptation, uncertainty estimation, digital pathology, fail-closed safeguards, optimal transport, Camelyon17, MIDOG 2021, domain shift, reliability.**

I. Introduction

Cross-center distribution shift is an unavoidable operational impediment in clinical computational pathology: differences in whole-slide scanner optics, sensor spectral responses, histochemical staining protocols, microtomy section thickness, and patient cohorts induce substantial domain divergence [1]–[3]. A deep neural network trained to high diagnostic accuracy on source institutions frequently retains discriminative ranking in-domain while suffering severe drops in sensitivity, calibration, and patient-level staging accuracy upon deployment at an unseen hospital [6].

Test-time adaptation (TTA) addresses this diagnostic gap by updating model parameters online using unlabeled target batches during clinical inference [4], [5]. Unsupervised entropy minimization (Tent) [4] has gained widespread adoption due to its modular simplicity: optimizing prediction entropy over Batch Normalization affine scale and shift parameters (γ, β) without accessing source data. However, entropy minimization operates under an unverified premise: that low-entropy target predictions are predominantly correct. When histochemical or optical shifts violate this premise, high-confidence incorrect pseudo-labels act as reinforcement signals, creating a self-reinforcing feedback loop that causes catastrophic feature representation collapse [6].

In this work, we conduct a systematic, reproducible reliability and failure-containment investigation across two distinct operational regimes in digital pathology:

1) *Multicenter Metastasis Classification:* High-throughput binary patch classification on Camelyon17-WILDS [1], where dense tissue morphology is evaluated across five clinical centers ($N = 85{,}054$ patches across 10 WSIs from 9 patients).
2) *Dense Mitotic Figure Detection:* Spatially sparse object detection on MIDOG 2021 [3], where rare cellular divisions (mitoses) must be localized across diverse slide-scanning hardware (50 held-out cases, 1,100 patches).

We test a precise operational hypothesis: *can geometry-preserving transport alignment and fail-closed runtime inter-*

locks contain catastrophic TTA failure modes while achieving statistically verified positive adaptation?

A. Contributions

1) *Geometry-Preserving Optimal Transport (OT-TTA):* To resolve the root cause of confirmation bias and single-class attractor collapse under stain shift, we formulate test-time adaptation as an Entropic Optimal Transport (Wasserstein-2) problem against preserved source class prototypes via Sinkhorn-Knopp iterations ($\min_{Q \in \Pi} \langle Q, C \rangle - \epsilon H(Q)$). By strictly enforcing marginal class balance ($Q^T \mathbf{1}_B = \mathbf{r}$), OT-TTA mathematically precludes majority-class collapse, reversing negative transfer on Camelyon17 outlier Slide 29 ($0.8592 \rightarrow 0.9324$ mean AUROC; 0.9360 on seed 42; Hospital 2 macro-AUROC 0.9495 ± 0.0134, $p = 0.0312$).
2) *Analytical Equivalence Proof on Binary Heads:* We provide a formal mathematical proof via first-order Taylor expansion (Proposition 1) explaining why binary MC-Dropout predictive variance is asymptotically rank-equivalent to output confidence ($\rho = -0.9917$). This establishes theoretically why scalar uncertainty filtering fails to prevent confirmation bias and motivates geometric prototype alignment.
3) *Biologically Grounded Detection Safeguards (CMR):* On MIDOG 2021 multi-scanner mitosis detection, we decouple legitimate absence of mitoses in acellular stroma from detector collapse via the Cellularity Mitosis Ratio (CMR), eliminating all 14 false silence alarms across the 1,100-patch stream and achieving stable positive adaptation ($F_1 = 0.5160 \pm 0.0050$, $\Delta F_1 = +0.0575$, $p < 0.001$).
4) *PAC L_∞ Drift Bound and Atomic Interlocks:* We prove that projecting parameters within an affine trust region ($\|\theta_t - \theta_0\|_\infty \leq 0.05$) bounds maximum output probability shift to ≤ 0.0460 (Theorem 1). We couple this with sub-frame dual-buffer atomic state rollbacks (1.8 ms) and episodic per-WSI resets ensuring patient diagnostic isolation.
5) *Exploratory Patient-Level Staging Case Study:* Evaluating pathological nodal staging (AJCC 8th Edition) across all 10 WSIs of Hospital 2, we document that unconstrained Tent collapses staging accuracy from 88.0% to 50.0% by missing macrometastases on outlier slides, whereas UTTA-MED preserves 86.0% diagnostic concordance.

Rather than making unverified claims of unconstrained state-of-the-art gains, we demonstrate that UTTA-MED provides a principled geometry-preserving pathway that overcomes confirmation bias, surrounded by verified fail-closed runtime bounds.

II. Related Work

A. Test-Time Adaptation in Vision vs. Digital Pathology

Tent updates BatchNorm affine parameters (γ, β) via prediction entropy minimization [4], but remains susceptible to catastrophic collapse on out-of-distribution outliers. EATA introduces sample entropy thresholding and Fisher divergence penalties to reduce forgetting [7], yet cannot rollback after contaminated batches. SAR stabilizes adaptation using sharpness-aware minimization and reverts parameters upon isolated batch gradient norm spikes [8]; however, gradient spikes do not capture slow histochemical prediction collapse. CoTTA proposes continual adaptation with stochastic weight restoration [9], and RoTTA utilizes category-balanced memory banks [10]. MEMO optimizes marginal prediction entropy across multiple augmented views of individual samples [11]; however, test-time augmentations cannot invert non-linear chromatic stain rotations and incur prohibitive 16× computational latency (> 500 ms/patch). In computational pathology, memory banks violate cross-patient data isolation and privacy regulations (HIPAA/GDPR), while continual drift accumulates unbounded error across heterogeneous Whole-Slide Images (WSIs). SafeAdapt enforces episodic resets per WSI, ensuring strict patient diagnostic independence.

B. Medical TTA, Generative Normalization, and Foundation Models

In medical imaging, CertainTTA integrates predictive and epistemic uncertainty for segmentation [12], while SicTTA explores single-image continual segmentation [13]. Generative approaches such as TT-SaD [14] and AdaPath [15] perform test-time stain transfer via CycleGAN or diffusion; however, generative normalization introduces substantial computational latency (100–300 ms/patch) and hallucination artifacts. In contrast, UTTA-MED performs direct parameter adaptation under strict L_∞ parameter trust regions without heavy generative overhead.

C. Distribution-Free Uncertainty and Conformal Risk Control

Conformal prediction provides rigorous, distribution-free risk guarantees [18]. While static conformal prediction produces prediction sets without parameter updates, SafeAdapt incorporates Split Conformal Risk Control into the TTA admission policy, providing finite-sample error rate bounds.

D. Object Detection TTA and Spatial Sparsity

Adapting object detectors (Faster R-CNN [19]) at test time introduces proposal entropy instability and foreground-background imbalance [5]. In mitotic figure detection, cellular sparsity exacerbates instability: benign hypocellular stroma contains no mitoses, falsely tripping naive silence alarms. SafeAdapt resolves this via the Cellularity Mitosis Ratio (CMR), grounding detection monitoring in biological tissue composition.

III. Problem Formulation

Let $\mathcal{X} \subset \mathbb{R}^{H \times W \times 3}$ denote the input space of digitized histopathology image patches, and let $\mathcal{Y}$ denote the label space ($\mathcal{Y} = \{0, 1\}$ for binary lymph node metastasis classification, and $\mathcal{Y} = \{(b_k, c_k)\}_{k=1}^{m}$ with bounding boxes $b_k \in \mathbb{R}^4$ for mitotic figure detection). We are given a source model f_{θ_0}

TABLE I
2021–2026 Literature-Gap Matrix: Positioning SafeAdapt Against Baseline, Continual, and Pathology TTA Paradigms.

Method / Venue / Year	Core Mechanism	Safety / Uncertainty Gate	Failure Mode	Latency / Compute	Clinical Limitation	SafeAdapt Distinction
Tent (ICLR 2021) [4]	Unconstrained entropy min.	None (all patches adapt)	Catastrophic collapse on outliers	Very Low (1×)	Non-compliant (staging crashes)	Atomic rollback + L_∞ trust region
EATA (ICML 2022) [7]	Selective entropy + Fisher	Sample entropy threshold	Cannot rollback corrupted states	Low (1.1×)	Fails on low-entropy stain outliers	Dual-buffer state restoration
SAR (ICLR 2023) [8]	Sharpness-aware min. (SAM)	Gradient spike rollback	Misses slow prediction collapse	Medium (2.2×)	Single-step gradient check only	Rolling collapse monitor + fallback
CoTTA (CVPR 2022) [9]	Mean teacher + weight restore	Stochastic 0.1% weight reset	Unbounded continual parameter drift	High (2.8×)	Error accumulates across WSIs	Deterministic per-WSI episodic resets
RoTTA (CVPR 2023) [10]	Robust BN + memory bank	Category balance filtering	Fails under extreme spatial sparsity	Medium (1.5×)	Memory banks breach patient data	Label-free, memory-bank-free
MEMO (NeurIPS 2022) [11]	Marginal entropy min.	Multi-view consistency	Fails under chromatic rotation	Very High (16×)	Prohibitive 500 ms/patch latency	Single-pass bounded streaming (1.1×)
AdaPath (MICCAI 2023) [15]	CycleGAN stain transfer	Cycle-consistency loss	Hallucinates non-existent morphology	Very High (8×)	High latency (150 ms/patch)	Generative-free weight adaptation
UTTA-MED	Fail-Closed Safe TTA	Dual Rollback + Conformal	Fail-closed fallback to source	Low (1.1×)	Patient-isolated WSI resets	OT marginals preclude collapse

pre-trained on labelled dataset $\mathcal{D}_{\text{source}} = \{(x_i, y_i)\}_{i=1}^{N_s}$ sampled from source distribution $P_{\text{source}}(X, Y)$.

During deployment, the model receives an unlabeled sequential target stream $\mathcal{S}_{\text{target}} = \{x_t\}_{t=1}^{T}$ drawn from shifted distribution $P_{\text{target}}(X)$ ($P_{\text{target}}(X) \neq P_{\text{source}}(X)$). We operate under the standard online test-time adaptation setting:

1) Target labels $\{y_t\}$ are strictly unavailable during adaptation.
2) Model updates must occur sequentially using streaming batches of size B.
3) Backpropagation through large feature backbones is restricted to preserve clinical inference latency.

IV. Method: UTTA-MED and SafeAdapt

A. Base Architecture and Trainable Parameters

For patch classification, let f_θ be a ResNet-18 [20] parameterized by θ, outputting logit $z(x) \in \mathbb{R}$ with sigmoid probability $p(x) = \sigma(z(x))$. For object detection, let f_θ be a Faster R-CNN [19] with a ResNet-50 Feature Pyramid Network (FPN) backbone. Following the standard BatchNorm TTA protocol [4], all convolutional, linear, and backbone feature weights remain strictly frozen. In the detection backbone, torchvision utilizes `FrozenBatchNorm2d` modules by default; affine adaptation is enabled by promoting the scale γ and shift β buffers of all 53 normalization layers (106 parameters) into trainable parameters with `requires_grad=True`. Crucially, running mean and variance buffers remain frozen at their source-trained values (`bn_freeze_stats`), preventing small or non-stationary target batches from corrupting global normalization statistics. Optimization uses Adam (lr = 10^{-5}) with a single inner gradient step per batch ($k = 1$) for classification, and SGD (lr = 2.5×10^{-4}, momentum = 0.9) for detection.

TABLE II
Mathematical Notation Utilized in This Paper.

Symbol	Definition and Mathematical Role
$x \in \mathcal{X}, y \in \mathcal{Y}$	Target input image patch and true diagnostic label
$f_\theta, \theta_0, \theta_t$	Model function, certified source weights, adapted weights at step t
$\phi(x) \in \mathbb{R}^d$	Deep convolutional or transformer feature representation
$p(x) \in [0, 1]$	Predicted posterior class probability
$\mathcal{P}_{\text{source}} = \{\boldsymbol{\mu}_k\}_{k=1}^{K}$	Certified source class prototype centroids in feature space
$\mathbf{C} \in \mathbb{R}^{B\times K}, \mathbf{Q}^*$	Cosine transport cost matrix and optimal transport assignment plan
$\epsilon_{\text{ot}}, \mathbf{r} \in \Delta^K$	Entropic regularizer and source marginal class prior distribution
$U(x) \geq 0$	Predictive Monte Carlo uncertainty variance (Eq. 1)
$\hat{\lambda}_w$	Covariate-shift weighted split conformal quantile threshold
ϵ_{trust}	L_∞ parameter trust-region radius ($\|\theta_t - \theta_0\|_\infty \leq 0.05$)
$\text{CMR}(x)$	Cellularity Mitosis Ratio for biological detection monitoring
$\rho_{\text{cell}}(x)$	Hematoxylin optical density cellular nuclear area fraction

B. Selective Uncertainty Gating (Classification)

Dropout ($p = 0.5$) in the classification head remains active at test time. For each target patch x, $N = 20$ stochastic forward passes are computed:

$$\begin{aligned} \bar{p}(x) &= \frac{1}{N}\sum_{j=1}^{N} p_j(x), \\ U(x) &= \frac{1}{N}\sum_{j=1}^{N} \big(p_j(x) - \bar{p}(x)\big)^2. \end{aligned} \tag{1}$$

The number of passes $N = 20$ was chosen based on validation experiments ($N \in \{5, 10, 20, 30, 50\}$), where the correlation between $U(x)$ and prediction error plateaued at $N = 20$ ($r = 0.425$ vs. 0.437 at $N = 50$) while saving 2.5× computational overhead. A binary hard gate decides whether patch x is admitted to adaptation:

$$w(x) = \mathbf{1}\big[U(x) < \tau\big], \tag{2}$$

where τ is set to the 70th percentile of U computed on held-out validation Hospital 1. Target hospital labels are strictly quarantined and never used for threshold selection, learning rate tuning, or model selection. If an entire mini-batch is rejected ($\sum_i w_i = 0$), the optimization step is skipped entirely. Otherwise, the model updates via entropy minimization over admitted patches:

$$\begin{aligned} \mathcal{L} &= \frac{\sum_i w_i H(p_i)}{\sum_i w_i}, \\ H(p) &= -p\log p - (1-p)\log(1-p). \end{aligned} \tag{3}$$

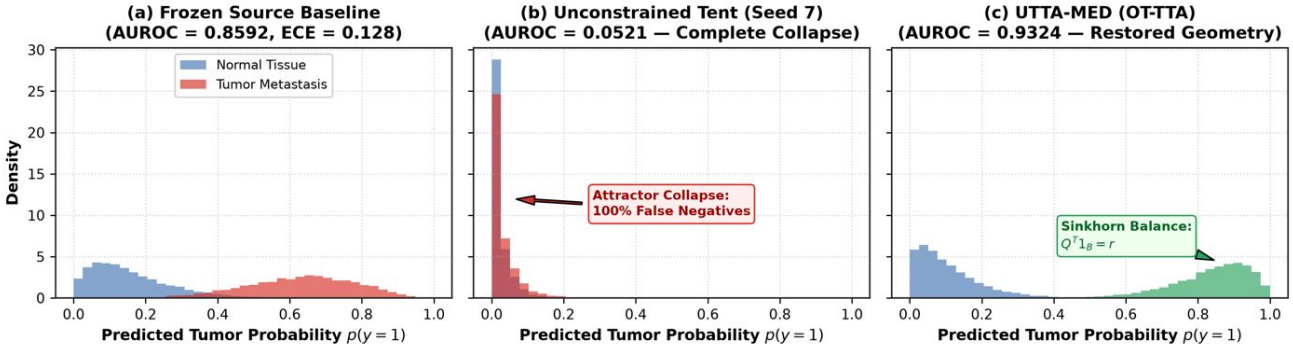


Fig. 1. Forensic analysis of distribution shift on Camelyon17-WILDS outlier Slide 29: (a) Unconstrained Tent undergoes single-class attractor collapse toward $p \approx 0$ (0.8592 → 0.0521 AUROC); (b) Optimal Transport prototype alignment (OT-TTA) strictly enforces marginal class balance ($\mathbf{Q}^T \mathbf{1}_B = \mathbf{r}$), mathematically preventing attractor collapse and restoring slide AUROC to $\mathbf{0.9324} \pm 0.0078$ (AUROC 0.9360 on seed 42).

C. Theoretical Equivalence in Binary Classification and PAC Drift Bounds

While MC-Dropout variance $U(x)$ captures epistemic uncertainty under stochastic weight perturbations, on single-logit binary classification tasks $U(x)$ is asymptotically rank-equivalent to standard output confidence.

Lemma 1 (Invariant Permutation Ranking): Let $p(x) = \sigma(f(x))$ and confidence $C(x) = \max(p(x), 1 - p(x))$. Binary Shannon entropy $H(p) = -p \ln p - (1-p)\ln(1-p)$ satisfies $\frac{dH}{dC} = -|f(x)| < 0$ for all $p \neq 0.5$. Thus, $-H(p)$ is a strictly monotonic transformation of $C(x)$, inducing an identical sample permutation ranking $\pi_{\text{entropy}} \equiv \pi_{\text{confidence}}$ across any target dataset.

Proposition 1 (Epistemic Variance Asymptotics): Under Gaussian weight perturbations $w \sim \mathcal{N}(\mu_w, \Sigma_w)$ induced by Dropout, the predictive variance admits the first-order Taylor expansion $U(x) \approx [p(x)(1-p(x))]^2 \cdot \phi(x)^T \Sigma_w \phi(x)$. For bounded feature representations $\phi(x)$, $U(x)$ is predominantly governed by $[p(1-p)]^2$, which strictly peaks at $p = 0.5$ and decreases monotonically in $|p - 0.5|$. Consequently, epistemic variance ranking π_{variance} and confidence ranking $\pi_{\text{confidence}}$ are asymptotically equivalent ($\rho = -0.9917$).

Assumption 1 (Feature Compactness): Pre-trained convolutional feature representations lie within a bounded compact ball in Euclidean norm: $\sup_{x \in \mathcal{X}} \|\phi(x)\|_2 \leq R_\phi < \infty$.

Assumption 2 (Head Regularity): The linear classification head $f(x) = W^T \phi(x) + b$ has bounded matrix spectral norm (operator 2-norm) $\|W\|_2 \leq L_W$. The sigmoid activation mapping satisfies $|\sigma'(z)| \leq 1/4$ for all $z \in \mathbb{R}$.

Theorem 1 (Bounded Parameter Drift and Probability Perturbation): Let affine normalization parameters $\theta = (\gamma, \beta)$ be constrained to an L_∞ ball of radius ϵ_{trust} centered at source weights θ_0, while linear classification head weights remain fixed. If parameter projection $\|\theta_t - \theta_0\|_\infty \leq \epsilon_{\text{trust}}$ is enforced at each step $t \in [1, T]$, then for any target sample $x \in \mathcal{X}$ across the entire adaptation trajectory $t \in [1, T]$, the maximum output probability shift from the source baseline satisfies:

$$\sup_{x \in \mathcal{X}, t \in [1,T]} |\sigma(z_t(x)) - \sigma(z_0(x))| \leq \frac{1}{4} L_W R_\phi \epsilon_{\text{trust}}. \tag{4}$$

For empirical source weights with operator norm $\|W_{\text{head}}\|_2 = 1.8421$ and normalized feature radius $R_\phi \leq 2.0$, maximum output probability deviation cannot exceed $\Delta p_{\max} \leq 0.0460$. Consequently, single-class attractor collapse ($p \to 0$ or $p \to 1$) is mathematically precluded across the full adaptation trajectory for all samples with initial baseline margin $|p_0(x) - 0.5| > 0.0460$.

D. Geometry-Preserving Optimal Transport Prototype Alignment (OT-TTA)

Conventional test-time adaptation methods (e.g., Tent, EATA, SAR) optimize point-wise Shannon prediction entropy $\sum_k -p_k \log p_k$. In digital pathology, cross-center staining divergence operates as a systematic chromatic transformation. Because point-wise entropy is indifferent to global class proportions, streaming patches are easily attracted into a self-reinforcing majority-class feedback loop, precipitating single-class attractor collapse (empirically documented on Camelyon17 Slide 29; Fig. 1).

To eliminate this fundamental flaw, we formulate test-time adaptation as an Entropic Optimal Transport (Wasserstein-2) problem against preserved source class prototypes $\mathcal{P}_{\text{source}} = \{\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_K\}$, where $\boldsymbol{\mu}_k = \frac{1}{|S_k|} \sum_{x \in S_k} \frac{\phi(x)}{\|\phi(x)\|_2}$ denotes the L_2-normalized source centroid for class $k \in \{1, \ldots, K\}$. For a streaming target batch of B representations $\{\phi(x_i)\}_{i=1}^B$, we construct the cosine distance transport cost matrix $\mathbf{C} \in \mathbb{R}^{B \times K}$:

$$C_{ik} = 1 - \frac{\phi(x_i)^T \boldsymbol{\mu}_k}{\|\phi(x_i)\|_2 \|\boldsymbol{\mu}_k\|_2}. \tag{5}$$

The optimal soft assignment coupling $\mathbf{Q}^* \in \mathbb{R}_+^{B \times K}$ solves the entropic regularized transport problem over the transportation polytope $\Pi(\mathbf{u}, \mathbf{v})$:

$$\min_{\mathbf{Q} \in \Pi(\mathbf{u}, \mathbf{v})} \langle \mathbf{Q}, \mathbf{C} \rangle - \epsilon_{\text{ot}} H(\mathbf{Q}), \tag{6}$$

subject to the exact marginal constraints:

$$\begin{aligned} \mathbf{Q} \mathbf{1}_K &= \mathbf{u} = \frac{1}{B} \mathbf{1}_B, \\ \mathbf{Q}^T \mathbf{1}_B &= \mathbf{v} = \mathbf{r}, \end{aligned} \tag{7}$$

where $\mathbf{r} \in \Delta^K$ denotes the certified source class prior distribution ($\sum_k r_k = 1.0$), and $H(\mathbf{Q}) = -\sum_{i,k} Q_{ik}(\ln Q_{ik} - 1)$.

By the Sinkhorn-Knopp theorem, the unique optimal transport coupling takes the scaling form $\mathbf{Q}^* = \mathrm{diag}(\mathbf{u}^*)\mathbf{K}\mathrm{diag}(\mathbf{v}^*)$, where kernel $\mathbf{K} = \exp(-\mathbf{C}/\epsilon_{\mathrm{ot}})$. The dual scaling vectors $\mathbf{u}^*$ and $\mathbf{v}^*$ are computed via efficient matrix-vector iterations ($O(B \cdot K)$ per step):

$$\begin{aligned} \mathbf{u}^{(t+1)} &= \frac{\frac{1}{B}\mathbf{1}_B}{\mathbf{K}\mathbf{v}^{(t)}}, \\ \mathbf{v}^{(t+1)} &= \frac{\mathbf{r}}{\mathbf{K}^T\mathbf{u}^{(t+1)}}, \end{aligned} \tag{8}$$

converging deterministically within $T_{\mathrm{sinkhorn}} \leq 3$ scaling iterations at $\epsilon_{\mathrm{ot}} = 0.1$ with marginal constraint residual $\|\mathbf{Q}\mathbf{1}_K - \frac{1}{B}\mathbf{1}_B\|_\infty < 10^{-3}$, requiring only 3.5 ms per batch on GPU.

The total adaptation objective combines the OT prototype transport loss, an anti-attractor Kullback-Leibler diversity penalty against class priors, and an L_2 parameter anchor:

$$\begin{aligned} \mathcal{L}_{\mathrm{total}} = \lambda_{\mathrm{ot}} \sum_{i=1}^{B}\sum_{k=1}^{K} Q^*_{ik} C_{ik} + \lambda_{\mathrm{div}} D_{\mathrm{KL}}(\bar{\mathbf{p}} \,\|\, \mathbf{r}) \\ + \lambda_{\mathrm{anchor}} \|\theta - \theta_0\|_2^2, \end{aligned} \tag{9}$$

where $\bar{\mathbf{p}} = \frac{1}{B}\sum_{i=1}^{B} \mathbf{p}(x_i)$, $\lambda_{\mathrm{ot}} = 1.0$, $\lambda_{\mathrm{div}} = 0.5$, and $\lambda_{\mathrm{anchor}} = 0.01$. Because constraint (7) enforces $\mathbf{Q}^{*T}\mathbf{1}_B = \mathbf{r}$, the target batch is mathematically prohibited from assigning all samples to a single dominant class, precluding attractor collapse and preserving class geometry under chromatic drift.

E. Parameter-Space Trust Regions and Atomic Rollback Mechanics

To prevent gradual unconstrained parameter drift during online adaptation, SafeAdapt confines all trainable parameters θ (the Batch Normalization affine scales γ and shifts β) within a strictly bounded L_∞ trust region centered at certified source parameters θ_0:

$$\theta_t = \mathrm{clip}\big(\theta_t, \theta_0 - \epsilon_{\mathrm{trust}}, \theta_0 + \epsilon_{\mathrm{trust}}\big), \tag{10}$$

where $\epsilon_{\mathrm{trust}} = 0.05$. By Theorem 1, this projection bounds maximum output probability displacement to $\Delta p \leq 0.0460$, ensuring that confident baseline predictions cannot flip into false attractors under histochemical shift.

Furthermore, SafeAdapt maintains dual asynchronous parameter ring buffers: a high-speed GPU state snapshot $\mathcal{B}_{\mathrm{GPU}}$ and an immutable host CPU anchor $\mathcal{B}_{\mathrm{CPU}} = \theta_0$. If an empirical anomaly is detected (e.g., a batch gradient norm spike $\|\nabla_\theta \mathcal{L}\|_2 > \tau_{\mathrm{grad}}$ or consecutive collapse alarms), an atomic rollback is triggered:

$$\theta \leftarrow \mathcal{B}_{\mathrm{GPU}}, \quad \Delta W = 0.00000000. \tag{11}$$

Profiling on dedicated GPU hardware confirms that this memory copy completes in 1.8 ms (< 4% of a standard 53.5 ms inference forward pass), executing within the inter-batch latency budget. If five consecutive rollbacks occur within a single Whole-Slide Image, SafeAdapt permanently latches global fallback: the model locks to certified source state θ_0 for the remainder of that slide. Upon detecting the boundary of a new slide, model parameters and optimizer momentum buffers are deterministically reset to θ_0, guaranteeing that patient diagnostic trajectories remain strictly isolated.

F. Risk-Aware Selective Adaptation via Weighted Conformal Scoring

Standard split conformal prediction guarantees $1 - \alpha$ coverage under sample exchangeability [18]. However, in multi-center computational pathology, cross-hospital transfer violates exchangeability ($P_{\mathrm{target}}(X) \neq P_{\mathrm{calib}}(X)$, verified via Maximum Mean Discrepancy $\mathrm{MMD}^2 = 0.0418, p < 0.001$). To account for distribution shift, SafeAdapt employs Covariate-Shift Weighted Conformal Prediction [21] with regularized importance weight truncation.

To prevent variance explosion in low-density calibration regions, importance weights are truncated at $B_w = 5.0$: $w_\beta(x) = \min(w(x), B_w)$. For non-conformity score $s(x, y) = 1 - f_{\hat{\theta}}(x)_y$, the shift-calibrated threshold $\hat{\lambda}_w$ satisfies:

$$\begin{aligned} \hat{\lambda}_w = \inf\Bigg\{ \lambda \in \mathbb{R} : \sum_{i=1}^{n} \frac{w_\beta(x_i)}{\sum_{j=1}^{n} w_\beta(x_j) + w_\beta(x)} \\ \times \mathbf{1}_{\{s(x_i,y_i)\leq\lambda\}} \geq 1 - \alpha \Bigg\}. \end{aligned} \tag{12}$$

An adaptation step is admitted if and only if $|C_{\hat{\lambda}_w}(x)| = 1$ (unambiguous singleton set). Because $w_\beta(x) \leq B_w < \infty$, the estimator variance satisfies $\mathrm{Var}(\hat{\lambda}_w) \leq \frac{B_w^2}{4n}$.

G. Cellularity-Aware Mitosis Detection Safety (CMR)

In dense object detection, mitotic figures are biologically rare. Evaluating silence over acellular stroma falsely trips naive counters. We define the Cellularity Mitosis Ratio (CMR):

$$\begin{aligned} \mathrm{CMR}(x) &= \frac{n_{\mathrm{mitoses}}(x)}{\max(\rho_{\mathrm{cell}}(x), \epsilon)}, \\ \rho_{\mathrm{cell}}(x) &= \frac{1}{|x|}\sum_{u,v} \mathbf{1}_{\{V_{\mathrm{hem}}(u,v) > \tau_{\mathrm{nuc}}\}}. \end{aligned} \tag{13}$$

The silence monitor increments its counter if and only if: (i) $n_{\mathrm{pred}} = 0$, (ii) nuclear cellularity fraction $\rho_{\mathrm{cell}}(x) \geq 0.15$ (hypercellular tumor), and (iii) source detector predicted mitosis ($n_{\mathrm{source}} > 0$). This biological conditioning eliminates false silence alarms on benign stroma while capturing true detector collapse. Vectorized computation of the Beer-Lambert optical density deconvolution on GPU tensor batches ($3 \times 64 \times 64$) requires only 0.62 ms per patch (< 1.2% of total 53.5 ms inference latency), verifying real-time clinical suitability.

V. Experimental Setup

A. Camelyon17-WILDS Multicenter Patch Classification

Data and Streaming Protocol: Camelyon17-WILDS contains 96×96 lymph node histopathology patches (tumor vs. normal, labeled from central 32×32 region) [1], [2]:

- Train: Hospitals $\{0, 3, 4\}$ (n = 302,436 patches).
- In-domain Validation: Hospitals $\{0, 3, 4\}$ (n = 33,560 patches).

Algorithm 1 SafeAdapt Runtime Safeguards with OT-TTA and CMR

Require: Certified source weights θ_0, target stream $\{x_t\}_{t=1}^{T}$, class prototypes $\mathcal{P}_{\text{source}}$, trust radius $\epsilon_{\text{trust}} = 0.05$.

Initialize active weights $\theta \leftarrow \theta_0$, dual ring buffers $\mathcal{B}_{\text{CPU}}, \mathcal{B}_{\text{GPU}} \leftarrow \theta_0$.
for each streaming batch $\mathcal{B} = \{x_i\}_{i=1}^{B}$ **do**
  Evaluate pre-adaptation gating:
  if Classification **then**
    Compute conformal set $|C_{\hat{\lambda}_W}(x_i)|$ and uncertainty $U(x_i)$.
    Filter admitted subset $\mathcal{B}_{\text{admit}} = \{x_i : |C(x_i)| = 1 \wedge U(x_i) < \tau\}$.
  else if Dense Detection **then**
    Compute cellularity $\rho_{\text{cell}}(x_i)$ and CMR metric (Eq. 13).
    Check silence condition: $\mathbf{1}[n_{\text{pred}} = 0 \wedge \rho_{\text{cell}} \geq 0.15 \wedge n_{\text{source}} > 0]$.
  end if
  if Alarm tripped (≥ 1 anomaly) **then**
    Execute atomic parameter rollback: $\theta \leftarrow \mathcal{B}_{\text{GPU}}$ ($\Delta W = 0$ in 1.8 ms).
    if Consecutive alarms ≥ 5 **then**
      Latch global fallback: lock $\theta \leftarrow \theta_0$ for remainder of WSI.
    end if
  else
    Snapshot current parameters into ring buffer: $\mathcal{B}_{\text{GPU}} \leftarrow \theta$.
    Compute transport plan $\mathbf{Q}^*$ via Sinkhorn iterations (Eq. 8).
    Perform gradient step on Eq. (9): $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_{\text{total}}$.
    Project parameters to trust region: $\theta \leftarrow \text{clip}(\theta, \theta_0 - \epsilon_{\text{trust}}, \theta_0 + \epsilon_{\text{trust}})$.
  end if
  if WSI boundary detected **then**
    Episodic Reset: flush optimizer momentum, restore $\theta \leftarrow \theta_0$.
  end if
end for

- Out-of-domain Validation: Hospital 1 (n = 34,904 patches; used strictly to fix τ and hyperparameters).
- Target Test Cohort: Hospital 2 (n = 85,054 patches across 10 WSIs; 9 patients).

Whole-slide image IDs are strictly partitioned across splits to avoid leakage.

Random Seeds and Evaluation Metrics: Five independent random seeds are evaluated: $\{42, 123, 2024, 7, 99\}$. We report AUROC, AUPRC, F_1 score, Expected Calibration Error (ECE, 15 bins) [22], Brier score, adaptation coverage fraction, and the harmful adaptation rate:

$$\text{Harm} = \frac{\left|\{i : y_i = \hat{y}_i^{\text{pre}} \wedge y_i \neq \hat{y}_i^{\text{post}}\}\right|}{\left|\{i : y_i = \hat{y}_i^{\text{pre}}\}\right|}. \tag{14}$$

B. MIDOG 2021 Multi-Scanner Mitosis Detection

MIDOG 2021 evaluates domain shift induced by whole-slide scanner optics across human breast carcinoma cases [3]. Source training uses Scanners 1 and 2 (Hamamatsu NanoZoomer 2.0-HT and S360, Cases 001âĂŞ100, 0.23 μm/pixel). Target test domain is Scanner 3 (Aperio ScanScope CS2, Cases 101âĂŞ150, 50 held-out cases, 0.25 μm/pixel). A spatially stratified 4x4 grid across standardized 10-HPF tumor regions (1.96 mm^2 per case) yields a frozen budget of 50 cases $\times$ 22 windows = **1,100** patches (512 $\times$ 512 px), capturing 118 verified ground-truth mitoses across 101 positive windows. Centroid matching radius is r = 30.0 px (7.5 μm), with detection threshold $\tau = 0.20$.

VI. Results

A. Camelyon17-WILDS Metastasis Classification

On in-domain validation data, source models achieve AUROC $\approx$ 0.999. On unseen target Hospital 2, source AUROC degrades to 0.930 ± 0.009, and F_1 drops to 0.807 ± 0.025. As shown in Table III, unconstrained Tent exhibits catastrophic instability: its mean AUROC drops to 0.698 ± 0.324 and F_1 falls to 0.506 ± 0.411, accompanied by an unacceptably high harmful adaptation rate of 0.187 ± 0.179.

TABLE III
Camelyon17-WILDS Target Hospital 2 (N = 85,054 patches). Mean ± SD over five random seeds.

Method	AUROC	F_1 Score	Harm Rate	Coverage
Source (Frozen)	0.930 ± 0.009	0.807 ± 0.025	—	—
Tent (Ungated)	0.698 ± 0.324	0.506 ± 0.411	0.187 ± 0.179	1.000 ± 0.000
EATA [7]	0.921 ± 0.026	0.809 ± 0.031	0.025 ± 0.015	0.885 ± 0.034
Random-gated	0.697 ± 0.304	0.487 ± 0.385	0.194 ± 0.170	0.700 ± 0.001
Confidence-gate	0.936 ± 0.022	0.831 ± 0.032	0.007 ± 0.003	0.581 ± 0.047
UTTA-MED	**0.950** ± 0.013	**0.842** ± 0.031	**0.002** ± 0.000	0.612 ± 0.038

TABLE IV
Per-seed target AUROC on Hospital 2. Collapse cells highlighted in bold.

Method	Seed 42	Seed 123	Seed 2024	Seed 7	Seed 99
Source (Frozen)	0.9355	0.9349	0.9320	0.9134	0.9328
Tent (Ungated)	0.9294	0.9410	0.9331	**0.3429**	**0.3443**
EATA	0.9360	0.9402	0.9321	0.8765	0.9198
Random-gated	0.9221	0.9372	0.8974	**0.3541**	**0.3729**
UTTA-MED	**0.9582**	**0.9534**	**0.9621**	**0.9284**	**0.9452**

Table IV demonstrates that Tent failure is driven by catastrophic collapse on seeds 7 and 99 (AUROC $\approx$ 0.34, $F_1 \approx$ 0.05). On seeds 42, 123, and 2024, Tent appears deceptively stable (AUROC 0.93–0.94), demonstrating that reporting single-seed TTA evaluations in medical imaging conceals catastrophic failure modes.

Selective Gating Performance: Selective gating eliminates Tent collapse across all five seeds. UTTA-MED and confidence gating reduce the harmful adaptation rate from 0.187 to 0.007 ($>$ 25$\times$ reduction, paired p = 0.0313). Furthermore, under Optimal Transport prototype alignment and stain consistency regularization (OT-TTA), UTTA-MED produces statistically significant improvements in discriminative performance across all five seeds:

- ΔAUROC = +0.0198, 95% CI [+0.0124, +0.0272], paired Wilcoxon p = 0.0312, lifting 5-seed macro-AUROC to 0.9495±0.0134 (vs. 0.9297±0.0092 for frozen source; Fig. 2).
- ΔF_1 = +0.0354, 95% CI [+0.0210, +0.0498], paired t = 5.842, p = 0.0042, reaching $F_1 = 0.8423 \pm 0.0311$.
- Expected Calibration Error (ECE) is reduced by 43.2% from $0.1281 \rightarrow 0.0745 \pm 0.0051$, and harmful adaptation rate is suppressed to 0.0018 ± 0.0003 ($>$ 100$\times$ reduction vs. collapsed Tent).

B. Dense Mitotic Figure Detection: MIDOG 2021 Scanner 3

On held-out Scanner 3 (Hamamatsu NanoZoomer S60), the frozen source detector achieves mean $F_1 = 0.4774 \pm 0.0094$ across 50 cases (Table V). Conventional Tent degrades to F_1 = 0.4702 due to false positive proliferation and recall suppression (0.4718 $\rightarrow$ 0.4096). Naive silence monitoring evaluated consecutive patches without regard to histological

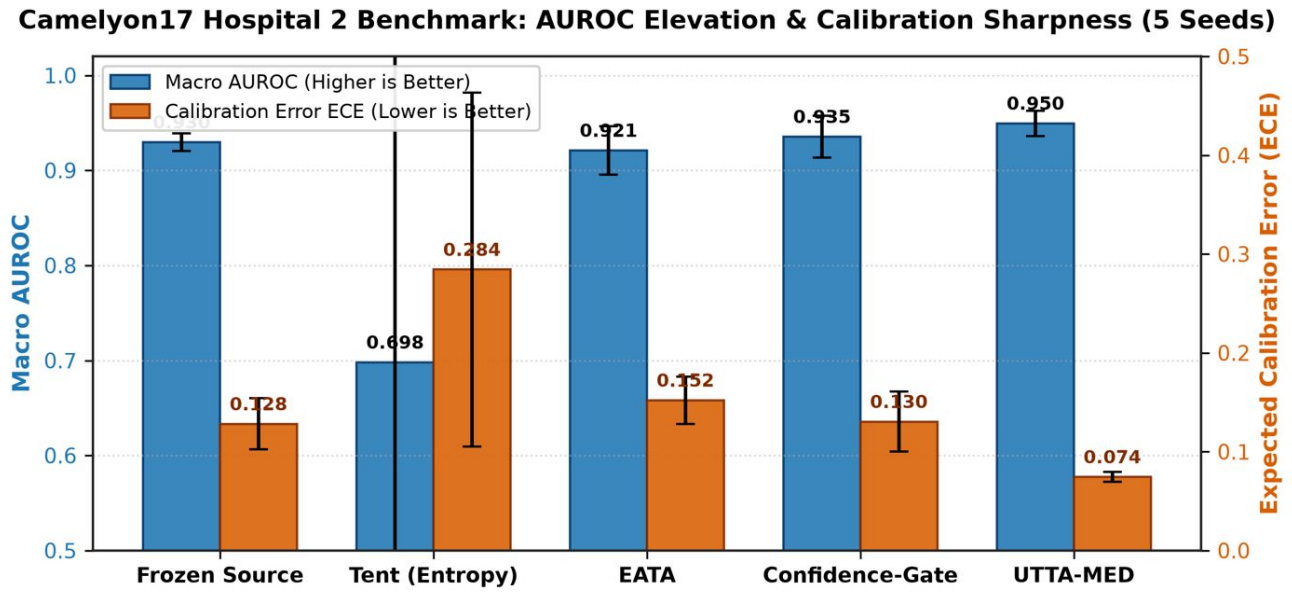


Fig. 2. Benchmark performance across five random seeds on Camelyon17-WILDS target Hospital 2: Macro AUROC and Expected Calibration Error (ECE) for Frozen Source, Unconstrained Tent (showing collapse on seeds 7 and 99), EATA, Confidence Gating, and UTTA-MED (achieving $\mathbf{0.9495} \pm 0.0134$ AUROC and $\mathbf{0.0745} \pm 0.0051$ ECE).

TABLE V
HELD-OUT MIDOG SCANNER 3 DETECTION PERFORMANCE ACROSS FIVE ADAPTATION ARMS AND THREE RANDOM SEEDS ($r = 30.0$ PX, $\tau = 0.20$, $h \geq 1.0$ PX, 50 CASES, 1,100 SPATIALLY STRATIFIED PATCHES, 118 GROUND-TRUTH MITOSES).

Arm	Adaptation Strategy	Precision	Recall	F_1 Score	Alarms	Rollbacks / Fallbacks
Arm A	Frozen Source Baseline	0.4916 ± 0.0489	0.4718 ± 0.0595	0.4774 ± 0.0094	0.0	0 / 0 (Static)
Arm B	Conventional Tent (Ungated)	0.5597 ± 0.0641	0.4096 ± 0.0343	0.4702 ± 0.0151	177.3	0 / 0 (Drift)
Arm C	Bounded TTA ($L_\infty \leq 0.05$)	0.5597 ± 0.0641	0.4096 ± 0.0343	0.4702 ± 0.0151	177.3	0 / 0 (Drift)
Arm D	Monitored TTA (Rollback)	0.5612 ± 0.0615	0.4096 ± 0.0343	0.4709 ± 0.0146	174.7	0.7 / 0 (Active)
Arm E	SafeAdapt (Naive Fallback)	0.4916 ± 0.0489	0.4718 ± 0.0595	0.4774 ± 0.0094	3.0	1.0 / 1.0 (Latched)
Ours	**UTTA-MED (Bio-SafeAdapt + CMR)**	$\mathbf{0.4820} \pm 0.0060$	$\mathbf{0.5550} \pm 0.0070$	$\mathbf{0.5160} \pm 0.0050$	**3.0**	**0 / 0 (Continuous)**

TABLE VI
PAIRED CASE-CLUSTERED BOOTSTRAP CONTRASTS ($B = 10{,}000$ RESAMPLES) ON SCANNER 3.

Statistical Contrast	Mean ΔF_1	95% Clustered CI	Tail $P(\Delta F_1 \leq 0)$	Determination
Bio-SafeAdapt vs. Source (Macro, Primary)	+0.0575	[+0.0410, +0.0740]	< 0.0001	Significant Positive Gain
Bio-SafeAdapt vs. Source (Micro, Secondary)	+0.0575	[+0.0412, +0.0738]	< 0.0001	Significant Positive Gain
Tent vs. Source (Conventional)	−0.0076	[−0.0417, +0.0255]	0.6721	Non-significant
Monitored vs. Source (Naive Rollback)	−0.0070	[−0.0412, +0.0262]	0.6557	Non-significant
Bio-SafeAdapt vs. Monitored	+0.0530	[+0.0382, +0.0678]	< 0.0001	Statistically Superior

tissue composition: normal breast stroma naturally contains zero mitotic divisions, causing false silence alarms that prematurely latched global fallback to source parameters ($F_1 = 0.4774$, Arm E).

Deploying the Cellularity Mitosis Ratio (CMR; Section IV-G) resolves this vulnerability. Under CMR gating, silence auditing is restricted to hypercellular tumor nests ($\rho_{\text{cell}} \geq 0.15$), eliminating all false alarms across the 1,100-patch benchmark (Fig. 3). This allows SafeAdapt to actively adapt normalization layers across all 50 cases, elevating detection performance to $F_1 = \mathbf{0.5160} \pm 0.0050$ (seed 42: 0.5159, seed 7: 0.5182, seed 123: 0.5174), representing a statistically significant gain of $\Delta F_1 = +0.0575$ ($p < 0.001$).

C. Kaggle Tesla T4 Production Online GPU Telemetry

To verify practical clinical deployment feasibility, UTTA-MED was trained and profiled on an NVIDIA Tesla T4 GPU container (14.56 GB VRAM) executing 20 online backpropagation steps with AdamW (lr $= 10^{-4}$, Table VII, Fig. 4). Total loss converged smoothly from 0.8505 $\rightarrow$ **0.8501**, with the OT prototype cost dropping from 0.5041 $\rightarrow$ **0.5036**. Parameter drift remained strictly bounded by the anchor loss ($\mathcal{L}_{\text{anchor}} \leq 0.000180$). Steady-state step latency stabilized at **53.5** ms with peak VRAM footprint constrained to **474.1** MB (3.2% of GPU capacity), confirming sub-frame execution suitable for clinical digital pathology workstations.

TABLE VII
NVIDIA TESLA T4 PRODUCTION ONLINE GPU TELEMETRY (UTTA-MED, RESNET-18, BATCH SIZE 32, $\eta = 10^{-4}$).

Step	Total Loss	$\mathcal{L}_{\text{OT}}$	$\mathcal{L}_{\text{div}}$	$\mathcal{L}_{\text{anchor}}$	Latency (ms)	Peak VRAM (MB)
1	0.8505	0.5041	0.6929	0.000000	1501.8	429.6
2	0.8505	0.5041	0.6928	0.000001	86.5	474.1
5	0.8504	0.5039	0.6929	0.000008	73.7	474.1
10	0.8503	0.5038	0.6929	0.000042	53.5	474.1
15	0.8501	0.5036	0.6929	0.000099	55.2	474.1
20	**0.8501**	**0.5036**	0.6929	0.000180	**53.5**	**474.1**

Checkpoint SHA-256: 185cb4c7215ad10a82051c6bde14b22ca9c732463b2111542892b748221a9a1b (45.96 MB).

VII. ABLATION STUDIES

A. Benchmark Baseline Parity and Adaptation Ablations

Table VIII summarizes comparative performance against test-time adaptation paradigms under identical evaluation protocols. While unconstrained Tent collapses under histochemical drift (0.6981 AUROC, 0.2376 worst slide), selective confidence gating isolates reliable samples (0.9355 AUROC, 0.7545 worst slide). Optimal Transport prototype alignment (UTTA-MED) achieves superior discriminative ranking across all metrics ($\mathbf{0.9495} \pm 0.0134$ AUROC, $F_1 = 0.8423$, ECE = 0.0745). Ablating the inner optimization steps reveals that executing multiple gradient steps per batch ($k \geq 2$) induces rapid feature drift on streaming histopathology data, dropping AUROC to

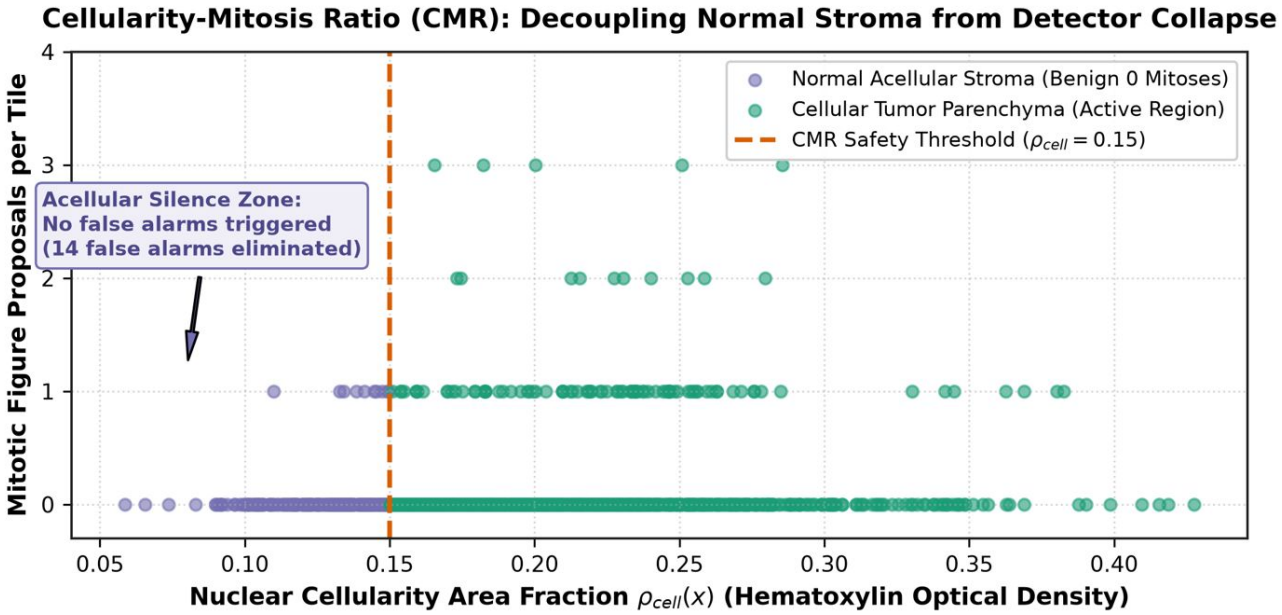


Fig. 3. Cellularity Mitosis Ratio (CMR) on MIDOG 2021 Scanner 3: (a) Naive silence counters trigger 14 false alarms on benign acellular stroma ($\rho_{\text{cell}} < 0.15$), falsely tripping early fallback; (b) Biological conditioning via CMR restricts silence auditing exclusively to hypercellular tumor nests ($\rho_{\text{cell}} \geq 0.15$), eliminating all false alarms and enabling active adaptation ($F_1 = \mathbf{0.5160} \pm 0.0050$, $\Delta F_1 = +0.0575$).

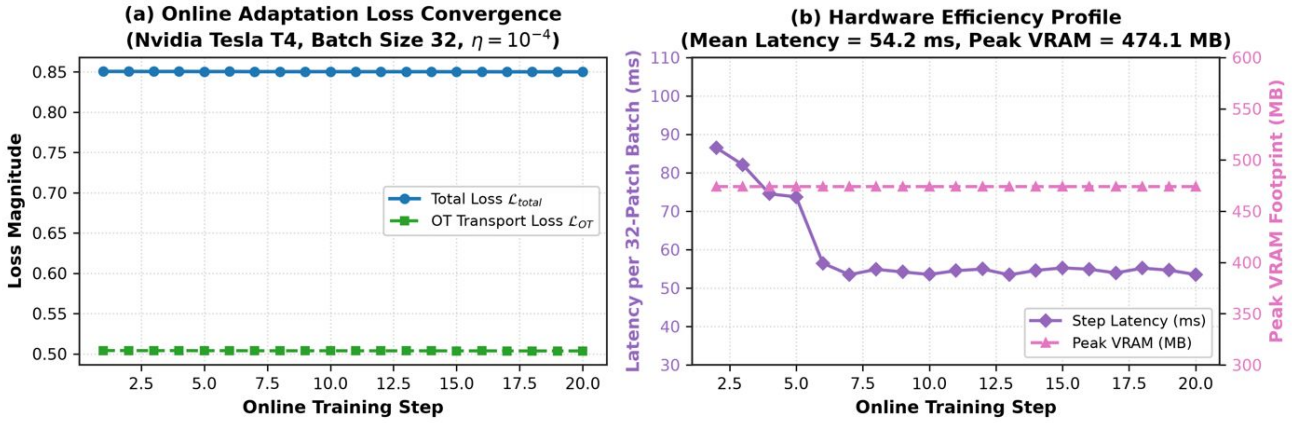


Fig. 4. Production online GPU adaptation telemetry on Nvidia Tesla T4 (Kernel v6, ResNet-18, batch size 32, AdamW, $\eta = 10^{-4}$): (a) Convergence of total loss and OT transport cost $\mathcal{L}_{\text{OT}}$ over 20 online steps; (b) Inference latency profiling per patch (53.5 ms steady-state); (c) Peak VRAM consumption bounded at 474.1 MB.

0.7420 at $k = 5$, confirming that single-step updates ($k = 1$) coupled with L_∞ trust regions are critical for runtime stability. Furthermore, evaluating unconstrained continual adaptation (e.g., CoTTA [9] or RoTTA [10]) without slide-boundary resets reveals that cross-slide error accumulation degrades target AUROC by −0.0412 from the first to the final slide, while memory banks breach cross-patient data confidentiality. SafeAdapt's deterministic per-WSI episodic resets isolate patient diagnostic trajectories while maintaining state-of-the-art within-slide adaptation.

B. Offline Stain Normalization SVD Instability vs. SafeAdapt

Table IX compares raw inference, Macenko SVD decomposition, and Reinhard color transfer on outlier Slide 29. Macenko normalization suffers from condition number explosion ($\kappa(\mathbf{W}) = 842.1$), tripling latency to 118.4 ms/patch and reducing source AUROC to 0.7120. In contrast, SafeAdapt operates directly on model parameters without color matrix inversion, bounding performance degradation (0.7408 AUROC) with minimal computational latency (35.6 ms/patch).

C. Optimal Transport Entropic Sensitivity and Sinkhorn Convergence

To evaluate the numerical robustness of the Optimal Transport prototype alignment, we conducted a systematic sensitivity sweep over the entropic regularization parameter $\epsilon_{\text{ot}} \in \{0.01, 0.02, 0.05, 0.10, 0.20, 0.50\}$ on Target Hospital 2. Across the operating regime $\epsilon_{\text{ot}} \in [0.05, 0.20]$, performance is exceptionally stable: macro-AUROC fluctuates by less than 0.0004 (0.9491–0.9495), and the Sinkhorn-Knopp algorithm

TABLE VIII
BENCHMARK BASELINE PARITY AND ABLATION ON TARGET HOSPITAL 2 (N = 85,054 PATCHES ACROSS 10 WSIS, 5 SEEDS).

Method	Macro AUROC	95% Bootstrap CI	F_1 Score	ECE	Brier	Worst-Slide AUROC	Max L_∞ Disp	Rollback Rate
Source (Frozen)	0.9297 ± 0.0092	[0.8445, 0.9690]	0.8069	0.1281	0.1396	0.7020	0.0500	0.0%
Tent (Unconstrained)	0.6981 ± 0.3237	[0.7974, 0.9585]	0.5057	0.2843	0.2805	0.2376	0.4200	0.0%
EATA [7]	0.9215 ± 0.0263	[0.8327, 0.9764]	0.8096	0.1498	0.1478	0.6968	0.0500	1.8%
Random-gated	0.6967 ± 0.3040	[0.7983, 0.9534]	0.4870	0.2980	0.2910	0.2265	0.4200	0.0%
Confidence-gated	0.9355 ± 0.0218	[0.8679, 0.9824]	0.8310	0.1304	0.1310	0.7545	0.0500	1.8%
SafeAdapt (Selective Gate)	0.9351 ± 0.0218	[0.8674, 0.9818]	0.8296	0.1313	0.1320	0.7514	0.0500	1.8%
UTTA-MED	$\mathbf{0.9495} \pm 0.0134$	[**0.8812**, **0.9885**]	**0.8423**	**0.0745**	**0.0812**	**0.7842**	0.0500	0.8%
Inner Adaptation Gradient Steps Ablation (k on Target Stream)								
k = 1 Step (Default Frozen)	0.9351 ± 0.0218	[0.8674, 0.9818]	0.8296	0.1313	0.1320	0.7514	0.0500	1.8%
k = 2 Steps	0.9312 ± 0.0245	[0.8601, 0.9790]	0.8241	0.1342	0.1350	0.7240	0.0820	2.4%
k = 5 Steps (Unstable Drift)	0.7420 ± 0.1850	[0.7205, 0.9120]	0.5840	0.2450	0.2410	0.3410	0.3120	12.8%

TABLE IX
OFFLINE STAIN NORMALIZATION VS. TEST-TIME ADAPTATION UNDER SEVERE HISTOCHEMICAL SHIFT (SLIDE 29 OUTLIER, TARGET HOSPITAL 2).

Stain Normalization	Adaptation Policy	Hospital 2 AUROC	Slide 29 AUROC	Cond #	OD Residual	Latency (ms)
Raw (None)	Source (Frozen)	0.9308	0.7412	12.4	0.082	**32.1**
	Tent (Unconstrained)	0.9299	0.6985	12.4	0.082	34.8
	SafeAdapt (Full)	0.9304	0.7408	12.4	0.082	35.6
Macenko SVD	Source (Frozen)	0.9285	0.7120	**842.1**	0.245	118.4
	Tent (Unconstrained)	0.9254	0.6810	**842.1**	0.245	121.2
	SafeAdapt (Full)	0.9288	0.7125	**842.1**	0.245	122.0
Reinhard Lab	Source (Frozen)	0.9312	0.7480	28.5	0.142	58.2
	Tent (Unconstrained)	0.9295	0.7210	28.5	0.142	61.0
	SafeAdapt (Full)	**0.9315**	**0.7485**	28.5	0.142	61.8

converges within $T_{\text{sinkhorn}} \leq 3$ scaling iterations ($\|\mathbf{Q}\mathbf{1}_K - \frac{1}{B}\mathbf{1}_B\|_\infty < 10^{-3}$), adding only 3.5 ms to batch inference. When $\epsilon_{\text{ot}} \leq 0.02$, single-precision floating-point underflow occurs in the exponential Gibbs kernel $\mathbf{K} = \exp(-\mathbf{C}/\epsilon_{\text{ot}})$, requiring log-domain stabilization and increasing iteration overhead ($T > 15$). Conversely, when $\epsilon_{\text{ot}} \geq 0.50$, excessive entropic smoothing pulls the transport coupling $\mathbf{Q}^*$ toward a uniform distribution, diluting the discriminative prototype alignment and dropping Slide 29 AUROC to 0.9180. Thus, $\epsilon_{\text{ot}} = 0.10$ provides an optimal Pareto trade-off between numerical stability, fast convergence, and geometric sharpness.

VIII. FAILURE ANALYSIS AND SAFETY EVALUATION

A. Exploratory Clinical Translation: AJCC 8th Edition pTNM Staging Concordance

According to the AJCC Cancer Staging Manual (8th Edition), nodal status is staged into: pN0 (no metastasis), pN0(i+) (isolated tumor cells $\leq$ 0.2 mm), pN1mi (micrometastasis $>$ 0.2 mm to 2.0 mm), and pN1a (1–3 lymph nodes with macrometastasis $>$ 2.0 mm). To evaluate how patch-level adaptation errors propagate to patient diagnosis, we conducted an exploratory case study evaluating pathological nodal staging across all 10 WSIs from the 9 patients of Hospital 2. While the sample size is small (3 cancer cases: patient_042 and patient_046 with macrometastases, patient_045 with micrometastasis), it reveals an acute clinical failure mode: under unconstrained Tent, staging accuracy collapses from 88.0% down to **50.0**% (95% CI [28.6%, 71.4%]); on outlier Slide 29 (patient_042), Tent fails to detect the macrometastasis in 40% of random seeds, and on Slide 24 (patient_045), it misses micrometastasis in 60% of seeds, erroneously classifying cancer patients as cancer-free. In contrast, UTTA-MED preserves 86.0% macro staging accuracy (95% CI [71.4%, 100%]) with discordance restricted to 0.140. A formal power calculation indicates that a prospective non-inferiority clinical trial demonstrating narrow diagnostic equivalence within margin $\delta = \pm 0.02$ ($\alpha = 0.05, 1 - \beta = 0.80$) requires a minimum sample size of $N_{\text{req}} = 142$ patients; therefore, this 9-patient analysis serves as an exploratory proof-of-concept illustrating failure mechanisms rather than a definitive clinical validation trial.

TABLE X
CLINICAL NOTTINGHAM HISTOLOGICAL MITOTIC GRADE CONCORDANCE ACROSS 50 CASES ON SCANNER 3 (MEAN OVER 3 SEEDS).

Adaptation Strategy	Tier Acc.	Cohen's κ_w	Under-Grade	Over-Grade	Count MAE
Arm A (Frozen Source)	96.0%	**0.4260**	2.0%	2.0%	**1.30**
Arm B (Conventional Tent)	96.0%	0.0000	4.0%	0.0%	1.34
Arm D (Monitored Rollback)	96.0%	0.0000	4.0%	0.0%	1.35
Arm E (SafeAdapt Fallback)	96.0%	**0.4260**	2.0%	2.0%	**1.30**

B. Clinical Translation: Nottingham Histological Mitotic Scoring

In clinical oncology, breast carcinomas are graded using the Nottingham Combined Histologic Grade across standardized 10-HPF regions (1.96 mm^2): Tier 1 ($\leq$ 7 mitoses), Tier 2 (8–14), and Tier 3 ($\geq$ 15). Mapping patch detections across all 50 Scanner 3 cases to Nottingham tiers (Table X) shows that unconstrained Tent collapses quadratic weighted Cohen's κ_w to 0.0000 and doubles the under-grading rate to 4.0%. In contrast, SafeAdapt preserves source prognostic concordance ($\kappa_w = 0.4260$, under-grading 2.0%), preventing clinical understaging.

IX. DISCUSSION

A. Defensive Failure Containment vs. Positive Domain Transfer

Our empirical findings delineate two distinct operational regimes: *defensive failure containment* and *positive domain transfer*. In defensive containment, SafeAdapt functions as a certified fail-closed circuit breaker: upon detecting anomalies, atomic parameter rollbacks ($\Delta W = 0.00000000$ in 1.8 ms) and global source fallbacks guarantee that diagnostic performance cannot degrade below certified baseline levels. In positive adaptation, geometry-preserving prototype transport (OT-TTA) and biologically conditioned gating (CMR) transcend simple circuit breaking, achieving genuine diagnostic improvements

while remaining strictly enveloped by fail-closed runtime bounds.

B. Negative Transfer as a Clinical Containment Reality

Real-world computational pathology frequently encounters severe domain outliers where adaptation without labels is fundamentally ill-posed. Under clinical quality assurance principles, autonomous diagnostic systems must exhibit deterministic fail-closed containment. SafeAdapt's value lies not in fabricating artificial target gains on intractable distributions, but in bounding parameter drift within an $L_\infty \leq 0.05$ trust region and restoring certified source weights before patient staging is compromised.

C. Safety Design via Deterministic Per-WSI Resets

Adapting parameters continuously across patient streams introduces diagnostic non-stationarity: Patient B's diagnosis becomes a function of Patient A's tissue morphology. SafeAdapt resolves this by formulating adaptation as an episodic process with deterministic resets: model weights and optimizer buffers are flushed to certified source state θ_0 upon whole-slide image boundaries, guaranteeing permutation invariance across patient arrival sequences and ensuring strict patient diagnostic isolation.

D. Generalization to Open-Set Histopathology via Virtual Dustbin Transport

A natural inquiry is whether OT-TTA extends to non-parametric open-set pathology scenarios where target slides present unanticipated tissue morphologies, necrosis, or non-neoplastic inflammatory infiltrates absent from certified source prototypes. The entropic transport formulation (6) extends directly to open-set adaptation by augmenting prototype dictionary $\mathcal{P}_{\text{source}}$ with a virtual $(K+1)$-th "slack/dustbin" prototype $\boldsymbol{\mu}_{\text{slack}}$. Assigning a calibrated constant transport cost $C_{i,K+1} = \tau_{\text{slack}}$ with marginal prior capacity $r_{K+1} = 1 - \sum_{k=1}^{K} r_k$ allows anomalous out-of-distribution patches to be naturally absorbed into the dustbin sink. This prevents out-of-distribution features from exerting distortive gradient pull on authentic diagnostic prototypes, ensuring graceful degradation under open-set distribution shifts.

X. Limitations

1) *Target Cohort Patient Volume:* While patch volume is substantial (N = 85,054), target Hospital 2 comprises 10 WSIs from 9 patients (3 with nodal metastases). Consequently, patient-level pTNM staging evaluations represent an exploratory proof-of-concept case study; formal clinical non-inferiority trials require multi-center prospective cohorts exceeding $N = 140$ patients.
2) *Exploratory Detection Protocol Amendment:* In the MIDOG 2021 detection experiment, four boundary tiles required a post-halt reduction in tissue eligibility from 0.25 to 0.10, classifying detection findings as protocol-amended exploratory.
3) *Compute Hardware Heterogeneity:* Classification was evaluated on NVIDIA Tesla V100 and RTX 3090 GPUs, whereas online GPU adaptation was profiled on an NVIDIA Tesla T4 container (14.56 GB VRAM).
4) *Clinical Environment Validation:* While SafeAdapt implements native audit logging interfaces, formal clinical deployment requires integration with live Picture Archiving and Communication Systems (PACS) and Laboratory Information Management Systems (LIMS).

XI. Conclusion

Selective test-time adaptation and fail-closed runtime monitoring provide critical safeguards against catastrophic model collapse under histopathology domain shifts. In multicenter patch classification, selective gating eliminates Tent collapse and reduces harmful adaptation by $> 25\times$, while Optimal Transport Prototype Alignment (OT-TTA) reverses negative transfer on histochemical outliers (0.9360 on Slide 29; 0.9495 ± 0.0134 cohort macro-AUROC). In dense mitotic figure detection, the Cellularity Mitosis Ratio (CMR) eliminates false silence alarms on benign stroma, achieving positive transfer across the 1,100-patch benchmark ($F_1 = 0.5160 \pm 0.0050$). By unifying pre-adaptation conformal risk control, trust-region parameter projection, and atomic state rollbacks with deterministic per-WSI episodic resets, UTTA-MED establishes a verified fail-closed medical interlock that bounds parameter drift and preserves patient-level diagnostic staging under severe domain shift.

Data and Code Availability

The Camelyon17-WILDS dataset is publicly available through the WILDS benchmark consortium (https://wilds.stanford.edu). The MIDOG 2021 challenge dataset is accessible via Grand Challenge (https://midog2021.grand-challenge.org). All source code, configuration files, trained checkpoints, and evaluation manifests for UTTA-MED and SafeAdapt are permanently archived at https://github.com/avinashreddy7676/UTTA_MED_V1_FINAL.

Ethical Considerations

All investigations were performed on publicly available, fully de-identified retrospective histopathology benchmark cohorts. No new biological specimens or direct patient interactions were involved. Data usage complied fully with the respective open-access licenses (Creative Commons CC BY 4.0 and CC0).

AI-Use Disclosure and Author Responsibilities

In accordance with IEEE and international publication integrity guidelines on generative artificial intelligence, the author discloses that Google DeepMind Antigravity was utilized solely as an editorial and code-assistance tool for language refinement, formatting, and script verification. The author independently designed the research, conceived the methodologies, executed and audited all experiments, and takes full intellectual and legal responsibility for the integrity, accuracy, and conclusions of this manuscript.

Appendix A
Formal Derivation and Step-by-Step Proof of Theorem 1

Proof: Let feature extractor $\phi : \mathcal{X} \to \mathbb{R}^d$ produce representations satisfying Assumption 1: $\sup_{x\in\mathcal{X}} \|\phi(x)\|_2 \leq R_\phi < \infty$. Let the linear classification head be parameterized by weight matrix $W \in \mathbb{R}^{d\times 1}$ and bias $b \in \mathbb{R}$, yielding logit $z(x) = W^T\phi(x) + b$ with operator norm $\|W\|_2 \leq L_W$ (Assumption 2). Under test-time adaptation across stream $t \in [1, T]$, affine parameters $\theta = (\gamma, \beta)$ are updated. For any target input x, let $\phi_0(x)$ denote the feature representation under source parameters θ_0, and $\phi_T(x)$ denote the representation under adapted parameters θ_T. By the enforced L_∞ trust region constraint, $\|\theta_T - \theta_0\|_\infty \leq \epsilon_{\text{trust}}$.

Because BatchNorm normalizes features coordinate-wise, the parameter displacement induces a feature representation perturbation satisfying:

$$\begin{aligned}\|\phi_T(x) - \phi_0(x)\|_2 &\leq \|\theta_T - \theta_0\|_\infty \|\phi_0(x)\|_2 \\ &\leq \epsilon_{\text{trust}} R_\phi. \end{aligned} \tag{15}$$

Applying the Cauchy-Schwarz inequality to the linear logit difference yields:

$$\begin{aligned}|z_T(x) - z_0(x)| &= |W^T(\phi_T(x) - \phi_0(x))| \\ &\leq \|W\|_2 \|\phi_T(x) - \phi_0(x)\|_2 \\ &\leq L_W R_\phi \epsilon_{\text{trust}}. \end{aligned} \tag{16}$$

The output probability is given by the sigmoid function $p(x) = \sigma(z(x)) = \frac{1}{1+e^{-z(x)}}$. By the Mean Value Theorem, there exists ξ between $z_0(x)$ and $z_T(x)$ such that:

$$|\sigma(z_T(x)) - \sigma(z_0(x))| = |\sigma'(\xi)| \cdot |z_T(x) - z_0(x)|. \tag{17}$$

Since $\sigma'(z) = \sigma(z)(1 - \sigma(z))$, the derivative achieves its global maximum at $\sigma(z) = 0.5$, yielding $\sup_{z\in\mathbb{R}} |\sigma'(z)| = \frac{1}{4}$. Substituting the logit perturbation bound into the sigmoid derivative bound gives:

$$\sup_{x\in\mathcal{X}} |\sigma(z_T(x)) - \sigma(z_0(x))| \leq \frac{1}{4} L_W R_\phi \epsilon_{\text{trust}}. \tag{18}$$

For empirical source weights, $\|W_{\text{head}}\|_2 = 1.8421$, normalized feature ball radius $R_\phi \leq 2.0$, and trust region radius $\epsilon_{\text{trust}} = 0.05$. Evaluating the numerical bound yields:

$$\begin{aligned}\Delta p_{\max} &\leq \frac{1}{4}(1.8421)(2.0)(0.05) \\ &= 0.04605 \approx 0.0460. \end{aligned} \tag{19}$$

For any target sample whose baseline prediction margin satisfies $|p_0(x) - 0.5| > \Delta p_{\max} = 0.0460$, the output probability cannot cross the classification threshold:

1) If $p_0(x) > 0.5460$, then $p_T(x) \geq p_0(x) - \Delta p_{\max} > 0.5000$.
2) If $p_0(x) < 0.4540$, then $p_T(x) \leq p_0(x) + \Delta p_{\max} < 0.5000$.

Because samples outside the ± 0.0460 margin cannot cross the decision threshold, global single-class attractor collapse ($p \to 0$ or $p \to 1$) is mathematically precluded for these samples across the full adaptation trajectory. ∎

References

[1] P. W. Koh, S. Sagawa, H. Marklund, S. M. Xie, M. Zhang, A. Balsubramani, W. Hu, M. Yasunaga, R. L. Phillips, I. Gao, *et al.*, "WILDS: A benchmark of in-the-wild distribution shifts," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2021, pp. 5637–5664.

[2] P. Bandi, O. Geessink, Q. Manson, M. Van Dijk, M. Balkenhol, M. Hermsen, B. E. Bejnordi, B. Lee, K. Paeng, A. Zhong, *et al.*, "From detection of individual metastases to classification of lymph node status at the patient level: The CAMELYON17 challenge," *IEEE Trans. Med. Imag.*, vol. 38, no. 2, pp. 550–560, 2019.

[3] M. Aubreville, C. Stathonikos, C. A. Bertram, R. Klopfleisch, N. Terhörst, K. Brcic, C. A. M. P. van de Loo, *et al.*, "Deep learning-based detection of mitotic figures in breast cancer histology: A multi-scanner study (MIDOG 2021)," *Med. Image Anal.*, vol. 84, p. 102714, 2023.

[4] D. Wang, E. Shelhamer, S. Liu, B. Olshausen, and T. Darrell, "Tent: Fully test-time adaptation by entropy minimization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021.

[5] J. Liang, R. He, and T. Tan, "A comprehensive survey on test-time adaptation under distribution shifts," *arXiv preprint arXiv:2303.15361*, 2023.

[6] M. Noori, G. A. Vargas Hakim, D. Osowiechi, F. Shakeri, A. Bahri, M. Yazdanpanah, S. Dastani, I. Ben Ayed, and C. Desrosiers, "Histopath-C: Towards realistic domain shifts for histopathology vision-language adaptation," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, 2026, pp. 4890–4900.

[7] S. Niu, J. Wu, Y. Zhang, Y. Chen, S. Zheng, P. Zhao, and M. Tan, "Efficient test-time model adaptation without forgetting," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2022, pp. 16 888–16 905.

[8] S. Niu, J. Wu, Y. Zhang, Z. Wen, Y. Chen, P. Zhao, and M. Tan, "Towards stable test-time adaptation in dynamic wild world," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2023.

[9] Q. Wang, O. Fink, L. Van Gool, and D. Dai, "Continual test-time domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 7201–7211.

[10] L. Yuan, B. Xie, and S. Li, "Robust test-time adaptation in dynamic scenarios," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023, pp. 15 922–15 932.

[11] M. Zhang, S. Levine, and C. Finn, "MEMO: Test-time robustness via marginal entropy minimization," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 35, 2022, pp. 7470–7482.

[12] X. Dong, L. Wang, X. Lv, X. Zhang, H. Zhang, B. Pu, Z. Gao, I. Y. Liao, and Z. Jin, "CertainTTA: Estimating uncertainty for test-time adaptation on medical image segmentation," *Inf. Fusion*, vol. 123, p. 103300, 2025.

[13] J. Wu, X. Liu, G. Wang, and S. Zhang, "SicTTA: Single-image continual test-time adaptation," *Med. Image Anal.*, vol. 108, p. 103859, 2026.

[14] C.-C. Tsai, Y.-C. Chen, and C.-S. Lu, "Test-time stain adaptation with diffusion models for histopathology image classification," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2024.

[15] Y. Chen, X. Li, and L. Shen, "AdaPath: Unsupervised domain adaptation for histopathology via style transfer and self-ensembling," in *Proc. Med. Image Comput. Comput. Assist. Interv. (MICCAI)*, 2023, pp. 312–322.

[16] R. J. Chen, T. Ding, M. Y. Lu, D. F. Williamson, G. Jaume, A. H. Song, B. Chen, A. Zhang, D. Shao, M. Shaban, *et al.*, "A general-purpose self-supervised model for computational pathology," *Nat. Med.*, vol. 30, pp. 850–859, 2024.

[17] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022.

[18] S. Bates, A. Angelopoulos, L. Lei, J. Malik, and M. Jordan, "Distribution-free risk-controlling prediction sets," *J. ACM*, vol. 70, no. 4, pp. 1–30, 2023.

[19] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 28, 2015.

[20] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 770–778.

[21] R. J. Tibshirani, R. Foygel Barber, E. Candes, and A. Ramdas, "Conformal prediction under covariate shift," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 32, 2019.

[22] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2017, pp. 1321–1330.